Assessing Implicit Attitudes Towards Wolfs Using the Affect Misattribution Procedure

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HUM-403 Experimental Cognitive Psychology

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21 December 2025

Assessing attitudes towards different animals using the affect misattribution procedure

When we look at an animal, we rarely see a blank slate. Instead, we see a collection of stories, myths, and emotions passed down through generations; narratives that often carry more weight than biological facts. Across societies, animals are frequently imbued with symbolic meanings that guide how they are evaluated, feared, or admired. These symbolic representations can persist over time and shape intuitive judgments, even when they conflict with empirical evidence. More broadly, research in affective processing suggests that we are not neutral observers. Instead, when we encounter something personally significant, most notably when it is perceived as a threat, our brains prioritize that information, biasing our perspective even with very brief exposure (Brosch et al., 2013). In the specific case of wolves, cultural representations have long linked the animal to danger and moralized notions of malevolence, contributing to the enduring "big bad wolf" stereotype (Jürgens & Hackett, 2017). Notably, this cultural image can persist despite contemporary ecological evidence indicating a continuing recovery of wolf populations across Europe, including their increasing presence in human-dominated landscapes, where public controversy often reflects social perceptions and fear more than objective risk (Di Bernardi et al., 2025). Psychological research has demonstrated that such culturally learned associations can operate at an implicit level. Implicit attitudes are automatic, rapid evaluations that occur outside conscious awareness and can influence judgments and behavior even when individuals explicitly reject stereotypical beliefs (Greenwald & Banaji, 1995). In the field, these perceptions translate into tangible consequences; as Mascia (2003) argues, human social and psychological factors are the primary forces shaping wildlife policy and the success of conservation strategies.

To investigate these spontaneous responses, researchers have turned to indirect measures that capture unfiltered affective triggers like the Affective Misattribution Procedure (AMP) (Payne et al., 2005), which is designed to isolate reflexive feelings. The

AMP assesses implicit affect by examining whether emotional responses elicited by a briefly presented prime influence subsequent evaluations of a neutral stimulus. Because the prime is presented too briefly to allow for deliberate correction, the task is well suited for measuring automatic affective responses. This paradigm has been extensively applied to the study of implicit evaluations across a wide range of domains, including social groups, moral concepts, and other affectively charged stimuli, and has demonstrated good reliability and construct validity across diverse experimental contexts (Deleuze et al., 2019)

Despite the broad application of implicit measures in social psychology, empirical research has rarely examined whether culturally constructed animal stereotypes manifest as measurable cognitive biases. While the humanities and social sciences have extensively documented the cultural association of wolves with danger and malevolence, psychological research on human–wolf relations has remained focused on explicit attitudes, demographic predictors, and socio-economic factors (Mascia, 2003). Consequently, a significant gap exists regarding whether these enduring cultural narratives produce automatic negative evaluations at an implicit level. Recent evidence suggests that these biases are malleable; for instance, Prokop et al. (2024) demonstrated that positive pedagogical interventions can significantly improve both the explicit and implicit attitudes of schoolchildren toward wolves. Addressing this gap is essential for advancing the theoretical understanding of implicit animal cognition and for developing evidence-based strategies for human–wildlife coexistence and conservation.

Original study

The current project replicates the experimental framework of Williams et al. (2016), which provided a critical empirical bridge between animal imagery and implicit response. Rather than relying on self-reports, their study utilized a child-friendly adaptation of the AMP to determine if animal-induced affect could be captured in its most raw form. By presenting unambiguously positive (e.g., puppies) or negative (e.g., sharks) primes before a neutral inkblot, the researchers measured the leakage of emotion. Since the rapid presentation of these images bypasses deliberate cognitive control, any shift in the evaluation of the neutral inkblot served as a proxy for the participant's spontaneous affective state.

The original results established a robust affective misattribution effect that was consistent across age and gender. Children did not merely observe the animal images; they spontaneously transferred the associated affect onto subsequent neutral stimuli. This finding is pivotal for the present research: it demonstrates that the AMP is sensitive enough to detect deep-seated emotional responses to animals. By extending this paradigm to include specific cultural archetypes, such as the wolf, we can move beyond "unambiguous" biological threats and examine whether socially transmitted narratives trigger the same reflexive evaluative biases.

Current study

The present study adapts the original paradigm to investigate whether the culturally entrenched stereotype of the "big bad wolf" can be measured as an implicit bias by transitioning the original child-based laboratory design to an online study with an adult population. The central goal is to determine if wolves elicit different emotional responses compared to physically similar animals that vary in their levels of domestication and familiarity. In this procedure, participants are briefly shown animal images before judging neutral stimuli, allowing for a measurement of how specific animals trigger automatic feelings that influence subsequent judgments.

To test these effects, the study uses three distinct categories: dogs, wolves, and foxes. These animals were selected to represent a spectrum of cultural and domestic associations. Dogs are expected to evoke predominantly positive affect due to their status as familiar domestic companions. In contrast, wolves are expected to elicit the most negative affect because of their long-standing cultural association with danger and malevolence. Foxes are included as a neutral comparison point, as they are wild animals that lack the extreme positive or negative stigmas attached to the other two groups. We hypothesize that dog primes will elicit the highest proportion of positive evaluations, wolf primes the lowest, and foxes will occupy an intermediate position between the two.

Material and Method

Participants

We plan to recruit 70 adult participants from the first-year psychology students at the University of Lausanne. Participation will take place as part of a credit-inducing program and no monetary compensation will be offered. Participation will be voluntary and students will be informed that they can refuse or withdraw at any time.

To be eligible, participants must be at least 18 years old, have a CEFR B2 level in English, and report normal or corrected-to-normal vision. We will exclude participants who do not complete the AMP task or who show clear signs of random responding (for example, extremely fast response times on most trials or failure to respond on a large proportion of trials). The final sample size and exclusion criteria will be reported in the Results report.

Material

The primes will be images of wolves, domestic dogs, foxes, and affective scenes. The images will be selected from online sources that allow non-commercial academic use (e.g., Unsplash or Pixabay) and will be standardized in size, resolution, and background as much as possible. We will avoid dogs that visually resemble wolves (e.g., huskies or German shepherds) and instead favor clearly domestic-looking breeds (e.g., beagles, dalmatians, Jack Russell terriers) to increase the perceptual distinction between dogs and wolves. To validate our AMP implementation, we will additionally use affectively normed images from the Open Affective Standardized Image Set (OASIS) (Kurdi et al., 2017). For each of the six prime categories (wolf, dog, fox, positive control, neutral control, negative control), we plan to select 15 images per category.

The neutral targets will be grayscale inkblots created by the experimenters. Each inkblot will be designed to avoid recognizable objects or faces and to have no obvious emotional content. Inkblots will be presented on a uniform background, centrally on the screen. A visual mask, a black-and-white noise pattern, will follow the target to interrupt further processing.

The experiment will be conducted online and accessed through a web link. Participants will complete the task remotely on their own computers. They will be instructed to perform the experiment in full-screen mode, in a quiet environment, and without interruptions. Responses will be collected via on-screen buttons labelled "pleasant" and "unpleasant," with their positions (left/right) kept constant throughout the task.

Procedure

Our procedure follows as closely as possible the procedure from the replicated study (Williams et al., 2016). After providing informed consent, participants will complete a short demographic questionnaire (age, gender) and a few explicit questions about their attitudes towards wolves, dogs, and foxes. These explicit measures are exploratory and will not be the primary focus of the present report. Participants will then receive instructions for the AMP, explaining that on each trial they will briefly see a picture (prime) followed by an inkblot, and that their task is to decide whether the inkblot looks "pleasant" or "unpleasant." The instructions will emphasize that the prime should not be taken into account for the rating and that there are no right or wrong answers. Participants will be encouraged to respond based on their first impression.

The task will begin with a short practice block of 10 trials where primes are shapes, to familiarize participants with the sequence of events and the response mapping. The main AMP task will then consist of all experimental and control primes. Not counting the 10 practice trials, each selected image will be presented once, resulting in 90 trials in total, divided into blocks of 15. Between each block, participants will have the possibility to take a break. The order of trials will be randomized separately for each participant, while ensuring each block contains both control and experimental images.

Each trial begins with a fixation cross at the center of the screen for 500 ms. This phase helps standardize gaze position and reduce variability in initial attention allocation

(Hughes et al., 2023). A blank screen will appear for 750 ms, followed by the prime image for 75 ms. After a 75 ms blank screen, the inkblot target will be displayed for 150 ms, immediately followed by the response screen (mask plus the labels “pleasant” and “unpleasant”). This screen will remain visible until the participant responds, after which the next trial starts. Brief feedback (e.g., “Too slow”) will be displayed when participants fail to respond within a reasonable time window.

Planned statistical analysis

For each trial, response data will be stored with each response indicating whether a target stimulus was judged as “pleasant” - coded as 1 - or “unpleasant” - coded as 0. - The data will be analyzed using a Generalized Linear Mixed Model (GLMM) with a binomial distribution and a logit link function, which is specifically suited for modeling binary responses (Singmann & Kellen, 2022) .

The main predictor of interest is the type of prime image presented before each target (wolf, dog, fox, and control categories). The analysis examines whether the category of the prime influences the likelihood of a pleasant response. Because all participants complete trials in every condition, the study uses a within-subjects design.

To account for the non-independence, participant will be included as a random effect in the model. Including random effects allows the analysis to take into account individual differences in overall response tendencies, such as a general bias toward pleasant or unpleasant judgments. This step is essential to avoid inflating statistical significance due to repeated measurements from the same participants.

The within-subjects design is particularly advantageous in this context because it allows direct comparison of responses to different prime categories while controlling for inter-individual variability. Each participant effectively serves as their own control, increasing statistical sensitivity.

The analysis will focus on comparing the proportion of pleasant responses elicited by different prime categories, particularly between dogs and wolves, as well as across dogs, foxes, and wolves to examine a potential domestication gradient. Statistical analyses will be implemented using JavaScript-based data processing and modeling tools. JavaScript has become a widely adopted language for implementing and analyzing behavioral experiments conducted online, allowing reliable data collection and processing directly in the browser (de Leeuw, 2015). Statistical significance will be evaluated using an alpha level of .05, which is the conventional threshold in psychological research and is appropriate given the exploratory yet theory-driven nature of the present study.

Transparency, openness, and reproducibility

The hypotheses and methods of the present study were not preregistered. However, the study design, materials, and analysis plan are described in sufficient detail to allow replication. Stimulus lists (including OASIS identifiers), procedures, and analysis scripts will be made available upon reasonable request. Anonymised trial-level data may also be shared, subject to ethical and legal constraints.

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